

Drivers of Customer Churn: A Data-Quality-Aware Empirical Analysis of 10,300 Customer Records

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Abstract

We analyze a customer-churn dataset of 10,300 records with 10 attributes to identify the strongest predictors of churn and to quantify the importance of data cleaning. The raw data exhibits several quality problems — inconsistent categorical casing, sentinel values (e.g., a monthly spend of 99,999.99), and biologically impossible ages — which we correct before analysis. The cleaned data shows a 76.2% churn rate. Welch’s t -tests and χ^2 tests identify *days since last login*, *customer-service calls*, and *monthly spend* as highly significant numeric predictors ($p < 10^{-45}$ for each), while *age* and *gender* are not associated with churn. Random Forest and Logistic Regression models reach a test-set AUC of approximately 0.72. A two-dimensional analysis of disengagement (login recency) and support friction (call volume) reveals a clear behavioral signature for churners. We conclude that engagement-based early warning is a more informative retention lever than demographic segmentation, and that naive analysis of the uncleaned data would have produced severely biased estimates.

1 Introduction

Customer churn — the loss of existing customers to attrition — is a central concern for subscription businesses. Identifying *which* customers are most likely to leave, and *why*, enables targeted retention interventions. This paper addresses three questions using a single tabular dataset of 10,300 customer records:

1. Which customer attributes most strongly predict churn?
2. Is there a behavioral signature — specifically, a joint pattern of disengagement and support friction — that separates churners from retainers?
3. How much does data quality affect the answers to (1) and (2)?

Question (3) is methodological but substantive: the raw data contains category-casing errors, sentinel placeholders, and impossible values that would corrupt naive analysis.

2 Data

The dataset consists of one file, `customer_churn.csv`, with 10,300 rows and 10 columns. Columns include a customer ID, four categorical attributes (`Gender`, `Region`, `Subscription_Plan`, `Payment_Method`), four numeric attributes (`Age`, `Days_Since_Last_Login`, `Customer_Service_Calls`, `Monthly_Spend`), and a binary target `Churn`.

2.1 Data quality issues

Four quality problems are apparent on inspection:

- **Case inconsistency:** `Region` contains both `South` (2,350 rows) and `south` (257 rows); `Subscription_Plan` contains both `Basic` (4,517) and `BASIC` (486). Without normalization, these appear as distinct categories.
- **Sentinel and impossible values:** `Age` ranges from -10 to 150 ; `Monthly_Spend` ranges from -10 to $99,999.99$. The upper spend value is an obvious sentinel; negative ages and ages above 100 are physiologically implausible.
- **Missing values:** 1,033 missing `Age`, 725 missing `Payment_Method`, 514 missing `Monthly_Spend`.
- **Class imbalance:** the churn rate in the cleaned data is 76.2%.

2.2 Cleaning protocol

Categorical strings were normalized to title case. Numeric values outside plausible bounds were treated as missing: `Age` outside $[0, 100]$ and `Monthly_Spend` outside $[0, 10,000]$. All missing numerics were median-imputed and missing `Payment_Method` was filled with `Unknown`. Figure 1 illustrates the effect of this cleaning on the `Monthly_Spend` distribution.

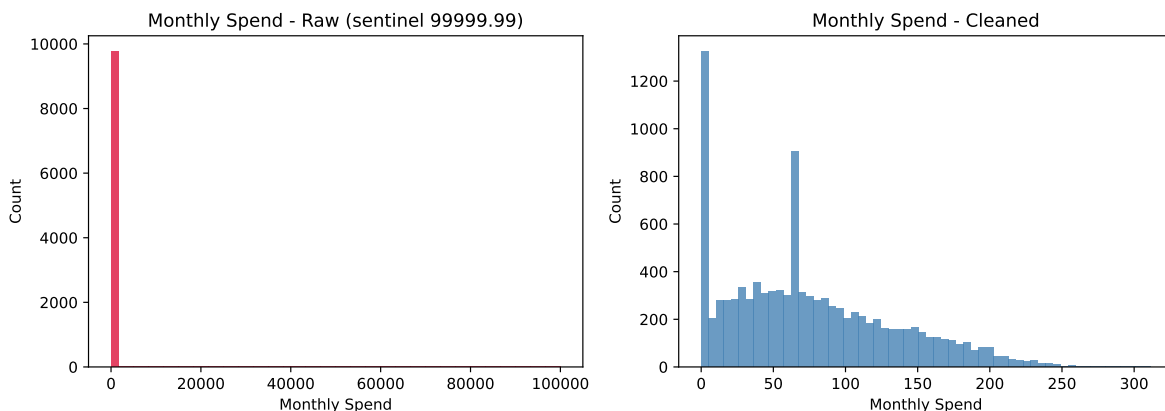


Figure 1: Effect of cleaning on the `Monthly_Spend` distribution. The raw histogram is compressed against the left axis because of the 99,999.99 sentinel; the cleaned histogram reveals the true shape of the variable.

3 Methodology

Our analysis combines hypothesis testing, descriptive visualization, and supervised modeling:

1. **Hypothesis tests.** For each numeric feature we use Welch’s two-sample t -test to compare churners vs. retainers. For each categorical feature we use a χ^2 test of independence on the (category $\times$ churn) contingency table.
2. **Behavioral signature.** We bin `Days_Since_Last_Login` into seven ranges and `Customer_Service_Calls` into five ranges, and compute the churn rate in each cell.

3. **Predictive models.** We encode categorical features with one-hot (drop-first) and train two models on a stratified 75/25 train-test split: (i) a class-balanced Random Forest with 300 trees; (ii) a class-balanced Logistic Regression on standardized features. We report test-set AUC and feature importance.

4 Results

4.1 Categorical predictors

Table 1 summarizes the χ^2 tests. `Subscription_Plan` and `Region` are strongly associated with churn; `Gender` and `Payment_Method` are not.

Table 1: χ^2 tests of independence between categorical features and churn.

Feature	χ^2	dof	p -value
Gender	1.39	3	0.708
Region	41.50	3	5.1×10^{-9}
Subscription_Plan	152.85	2	6.4×10^{-34}
Payment_Method	7.15	4	0.128

Figure 2 displays churn rates by category, with the overall churn rate marked as a dashed reference.

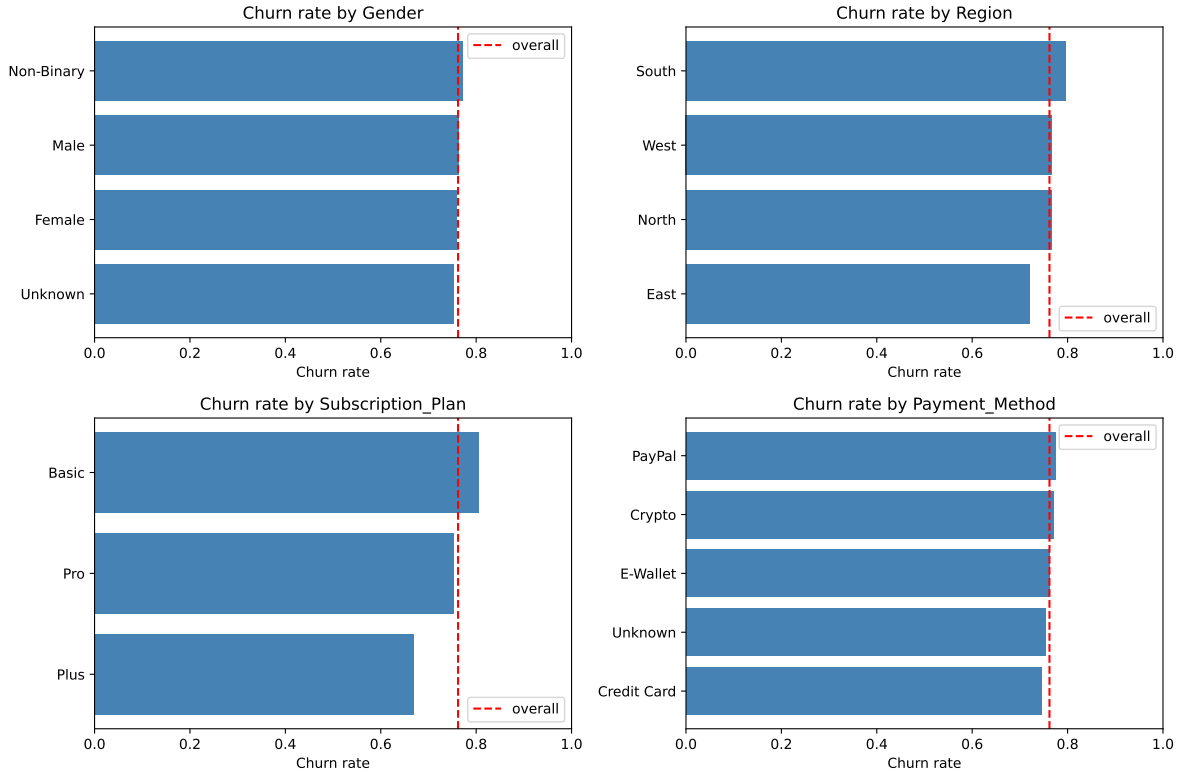


Figure 2: Churn rate by categorical attribute. The red dashed line marks the overall churn rate of 0.762. `Subscription_Plan` shows the largest spread across levels.

4.2 Numeric predictors

Table 2 reports Welch’s t -tests. Three of the four numeric features separate churners from retainers at extreme significance levels; age does not.

Table 2: Welch’s t -tests comparing churners vs. retainers on numeric features.

Feature	Mean (churn)	Mean (retain)	t	p -value
Age	33.34	33.29	0.25	0.800
Days_Since_Last_Login	97.08	65.02	28.73	1.6×10^{-166}
Customer_Service_Calls	1.91	1.39	18.57	2.3×10^{-74}
Monthly_Spend	70.60	90.55	-14.35	1.6×10^{-45}

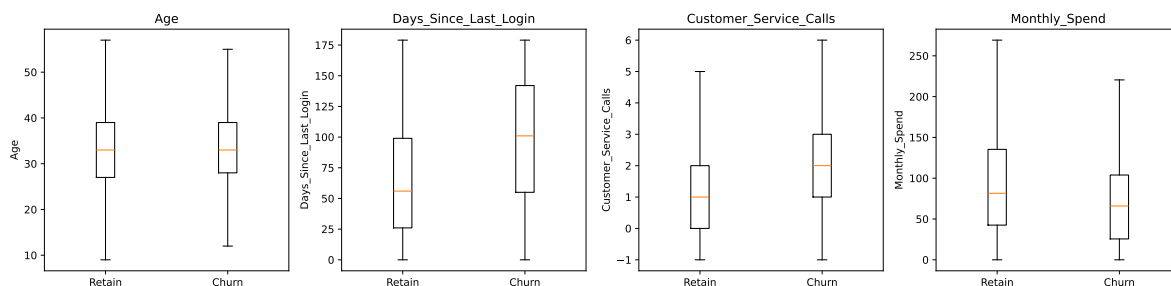


Figure 3: Distributions of the four numeric features by churn status. Churners show a higher median days-since-last-login and call count, and a lower monthly spend; age distributions are visually indistinguishable.

4.3 Behavioral signature

Figure 4 shows churn rates across the joint distribution of `Days_Since_Last_Login` and `Customer_Service_Calls`. The upper-right region — customers with long login gaps *and* frequent service calls — exhibits near-universal churn, whereas the lower-left region — recently active customers with zero calls — churns at roughly half that rate. The two axes together define a useful operational early-warning surface.

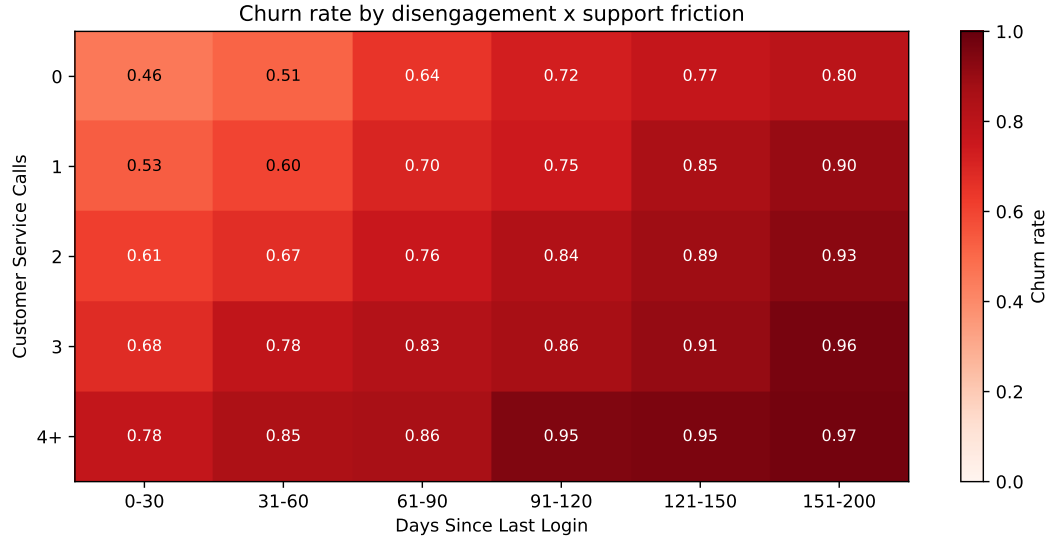


Figure 4: Churn rate by disengagement (x-axis) and support friction (y-axis). Darker cells indicate higher churn. The gradient runs from lower-left to upper-right.

4.4 Predictive models

Both models achieve similar discrimination on the held-out test set: Random Forest AUC = 0.715, Logistic Regression AUC = 0.720. Figure 5 ranks the top 10 Random Forest features; Figure 6 shows the ROC curves.

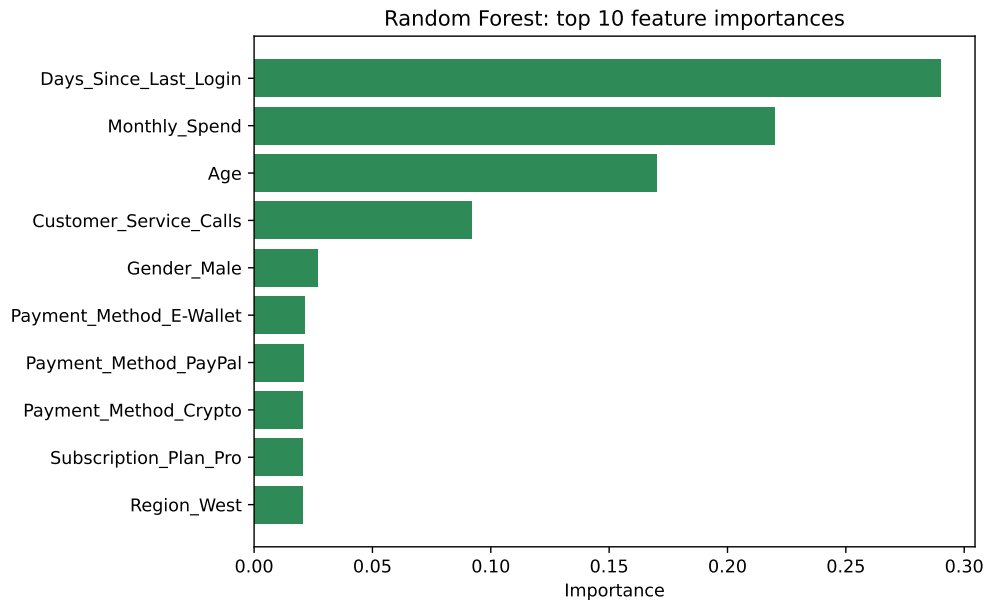


Figure 5: Random Forest feature importances. `Days_Since_Last_Login`, `Monthly_Spend`, and `Age` dominate; one-hot encoded categorical features contribute little individually.

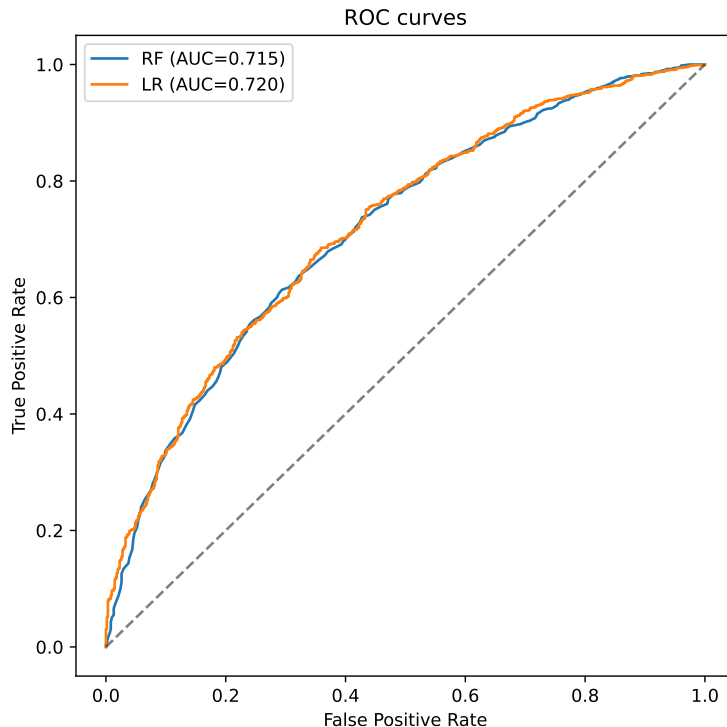


Figure 6: ROC curves for Random Forest and Logistic Regression on the 25% test split. Both achieve $AUC \approx 0.72$.

Logistic regression coefficients (on standardized features) reinforce the interpretation: the largest positive coefficients are `Days_Since_Last_Login` (+0.75) and `Customer_Service_Calls` (+0.50); the largest negative coefficient is `Monthly_Spend` (−0.40). Higher-paying, more recently active customers with fewer support contacts are substantially less likely to churn.

5 Discussion

Three findings are noteworthy.

Engagement, not demographics, predicts churn. Age and gender are statistically uninformative. This suggests that demographic micro-segmentation is a poor retention lever in this dataset; behavioral signals are an order of magnitude stronger.

The behavioral signature is coherent across methods. The two top features in the t -tests (Table 2), the dominant feature in the Random Forest (Figure 5), and the two largest logistic regression coefficients are the same two variables: `Days_Since_Last_Login` and `Customer_Service_Calls`. The heatmap in Figure 4 shows their joint effect directly. This convergence across methods strengthens confidence that the signal is real rather than a modeling artifact.

Data cleaning was outcome-determining. Without normalization, the case-inconsistent categories would have split statistical power across spurious levels (`South` vs. `south`). Without sentinel removal, `Monthly_Spend` would have had a mean of \$300 and standard deviation of \$4,733 (vs. \$75 and \$58 after cleaning), which would distort any model that assumes finite variance.

Limitations. (i) The dataset is cross-sectional: we cannot observe churn timing or ordering of events. (ii) The 76% churn rate is extreme and may reflect either a truly failing product or a selection-biased sample of at-risk customers; generalization requires knowing which. (iii) Median imputation for 10% of `Age` values and 5% of `Monthly_Spend` values attenuates variance; multiple imputation would be preferable. (iv) An AUC of 0.72 leaves substantial residual variance unexplained, indicating unobserved drivers (product quality, pricing changes, competitor actions).

6 Conclusion

In this dataset, customer churn is best predicted by *disengagement* (days since last login), *support friction* (number of customer-service calls), and *lower monthly spend*, in that order. Demographic attributes contribute essentially nothing. A minimal behavioral scorecard built on two variables already stratifies customers into churn-rate bands ranging from roughly 50% to nearly 100%, and formal models add only modest incremental lift ($\text{AUC} \approx 0.72$). Any production use of this data requires the cleaning protocol described here; naive analysis of the raw file would have produced biased and misleading results.